

Acceptance Test Case Documentation

Test Case Title: Handling Multiple Elevator Requests With Capacity Limits, Retries, and Faults

Objective:

To verify the complete system functionality under high-load conditions with multiple elevators, capacity limits, retry logic, and injected faults — satisfying the acceptance testing requirement for SYSC 3303.

Test Setup

- **System Configuration:**
 - 4 Elevators running concurrently (Elevator 1 to Elevator 4)
 - Each elevator has a capacity of 4 passengers
 - Heartbeat timeout: 15 seconds
 - Retry logic enabled when elevator is full
 - Fault flags:
 - 0: Normal
 - 1: Transient fault (e.g., door stuck)
 - 2: Hard fault (elevator crash)
- **Input File:** Timestamped request list, Max floor: 22, Min floor: 0

Sample Input Breakdown (selected highlights)

Time	Floor	Direction	Destination	Fault
14:00:00.0	1	Up	5	0
14:00:00.0	1	Up	6	2 ⚠ (hard fault)
14:00:00.0	1	Up	9	1 ⚠ (transient)
14:00:10.0	2	Down	1	0

Total requests: **26**

Expected Behaviors

Feature	Expected Result
Elevator Dispatch	All requests from Floor 1 are assigned to multiple elevators to balance load.
Capacity Management	Once an elevator reaches capacity (4/4), it sends remaining request(s) back to Scheduler for reassignment.
Retry Logic	Requests that couldn't be boarded are retried and picked up by the assigned elevator.
Multi-elevator Coordination	Floor 1 requests should be distributed among Elevators 1–4.
Transient Faults	Transient fault at Floor 9 should cause delay, retries, and recovery.
Hard Faults	Fault flag 2 for Floor 6 should result in shutdown of that elevator. No further requests should be assigned to it.
Final Cleanup	All valid requests should eventually be served, unless all elevators become unavailable due to faults or shutdown. The system should continue operating with available elevators and must not crash.

Success Criteria

- Each request is acknowledged in the Floor logs (i.e., picked up and dropped off).
- Retry messages for capacity overflow are seen (e.g., "Button pressed again...").
- Elevator logs indicate response to fault injection.
- System handles faults gracefully, no crashes.
- Elevators that shut down are skipped in assignment.
- Scheduler logs show retries being reprocessed.